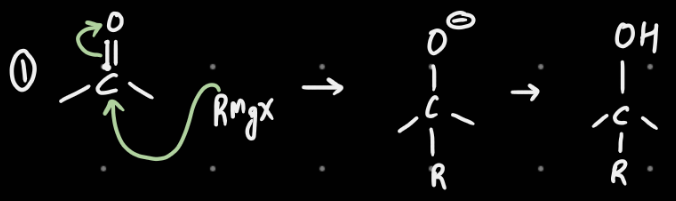
ALCOHOLS

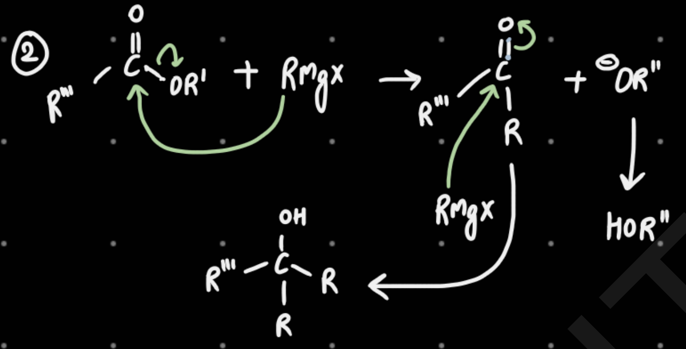
(I) Prep:

1. Grignard

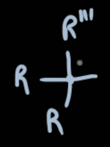
①


O
RMgX
O⊖
R
OH
R

②

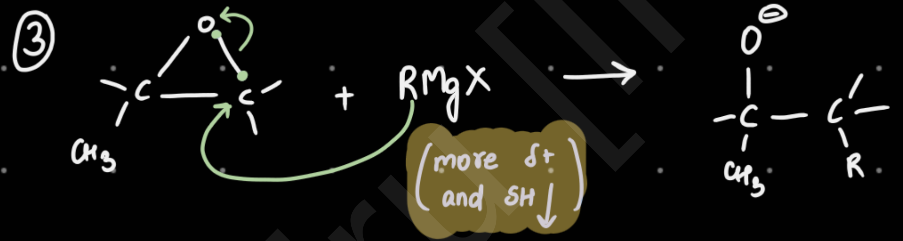

O
R'''
OR'
+ RMgX
R'''
O
R
+ ⊖OR''
HOR''
RMgX
OH
R'''
R
R

{ For single alc.
↳ $R'' =$



}

③


O
CH_3
+ RMgX
O⊖
CH_3
R

(more δ+ and SH↓)

2. Reduction

(A) R.A ⇒ a. $LiAlH_4$ / Ether → Source of $H^\ominus$

b. $NaBH_4$ / EtOH

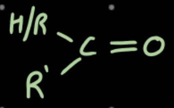
c. $NaBH_4$ + $AlCl_3$ in dry glyme

d. B_2H_6 / THF

e. H_2 / Catalyst

Li Na
Nahi
Bachhi

① Aldehyde / ketone → a b c d e



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② Acid / Acid Anhydride $\longrightarrow$ a ~~b~~ c d e → only Cu-Ba-Cr oxide

$R-\overset{O}{\overset{\|}{C}}-OH$ $\qquad R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$

③ Ester $R-\overset{O}{\overset{\|}{C}}-OR'$ $\longrightarrow$ a b ~~c~~ d e

↳ Excess of $NaBH_4$ in MeOH.

④ Acid Chloride $R-\overset{O}{\overset{\|}{C}}-Cl$ $\longrightarrow$ a b c ~~d~~ e

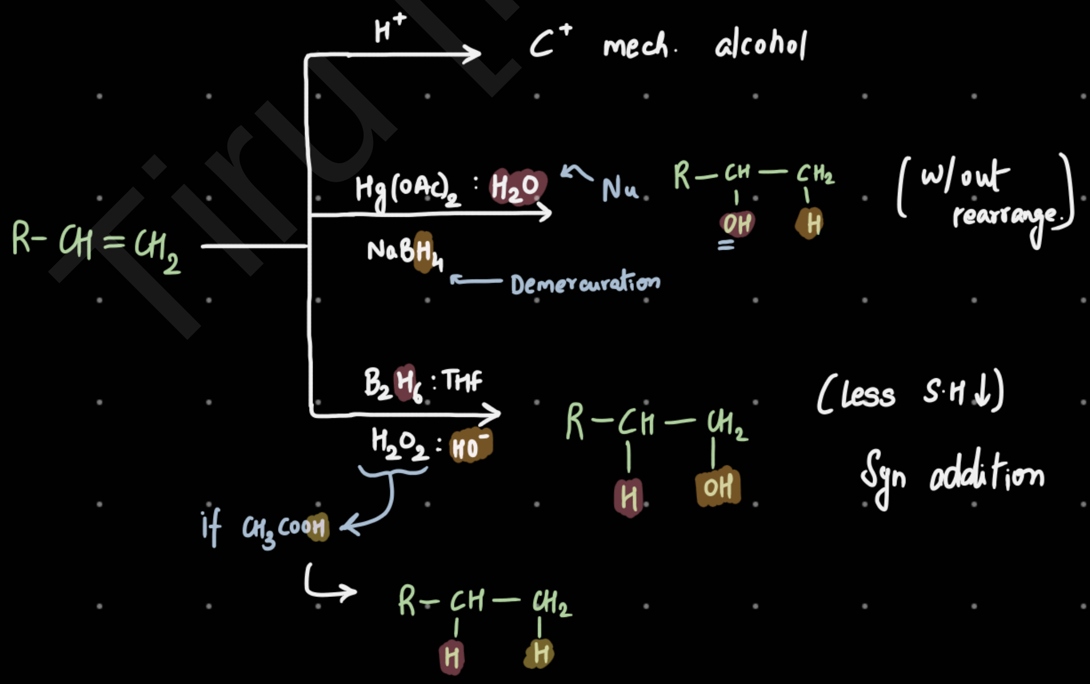
④ Alkene $>C=C<$ $\longrightarrow$ ~~a~~ ~~b~~ c d e

⑤ Alkyne $-C\equiv C-$ $\longrightarrow$ ~~a~~ ~~b~~ ~~c~~ d e

(B)

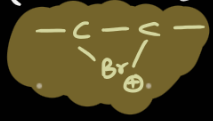
RA	$C=O$	$C=C$
$LiAlH_4$	✓	× → (conjugated at T↑) ✓
9-BBN	✓	×
$NaBH_4$	✓	50%
H_2 / cat.	✓	✓

3. Hydration


$R-CH=CH_2$
H^+
C^+ mech. alcohol
$Hg(OAc)_2$: H_2O
Nu
$NaBH_4$
Demercuration
$R-CH-CH_2$
OH
H
(w/out rearrange.)
B_2H_6 : THF
H_2O_2 : HO^-
$R-CH-CH_2$
H
OH
(Less S.H↓)
Syn addition
if CH_3COOH
$R-CH-CH_2$
H
H

* Alkene + Br_2 + H_2O $\longrightarrow$ Halohydrin

(no rearr.)



(II) Tests

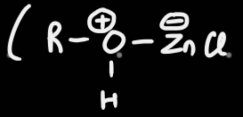
1. Lucas Test

conc. HCl + $ZnCl_2$

$3° > 2° > 1°$

S_N1 S_N1 S_N2

If 2° ⇒ S_N2
↳ $3° > 1° > 2°$



$$\left(R-\underset{H}{\overset{\oplus}{O}}-\overset{\ominus}{Zn}Cl \quad \text{soluble in HCl} \right)$$

2. Victor Meyer Test

PIANO Nahi Na !

Steps : $(P + I_2) \rightarrow (AgNO_2) \rightarrow (NaNO_2 + HCl) \rightarrow (NaOH)$

1° 2° 3°
R B C
Red Blue Colourless

3. Iodoform Test

↳ $NaOH + I_2$ / $NaCO_3 + I_2$

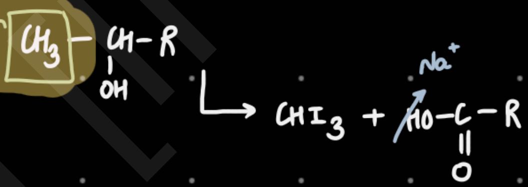
yellow ppt of $\{CHI_3\}$

$CHCl_3$ $CHBr_3$
colourless

1° → ✓

2° → At least one

3° → ×××

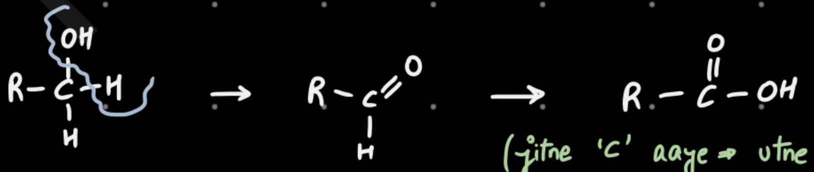


$$\boxed{CH_3}-\underset{OH}{CH}-R \;\rightarrow\; CHI_3 + \overset{Na^+}{\cancel{H}}O-\underset{\underset{O}{\|}}{C}-R$$

(III) Oxidation

O.A → Mn(VII) , Cr(VI)

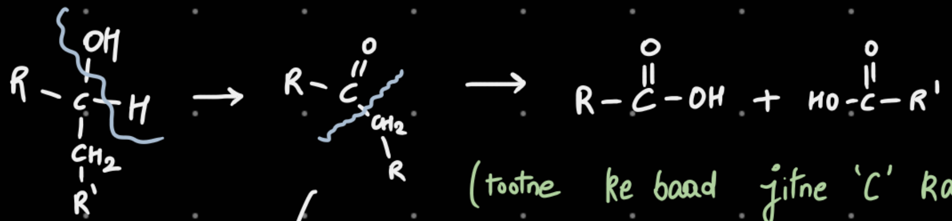
1° ⇒



$$R-\underset{H}{\overset{OH}{C}}-H \rightarrow R-\underset{H}{C}=O \rightarrow R-\overset{O}{\overset{\|}{C}}-OH$$

(jitne 'C' aaye ⇒ utne C ka acid banado)

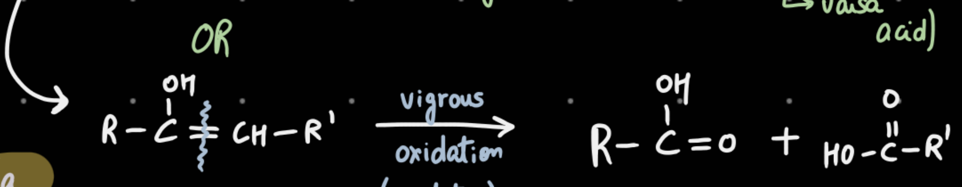
2° ⇒



$$R-\underset{\underset{R'}{|}}{\underset{CH_2}{\overset{OH}{C}}}-H \rightarrow R-\overset{O}{\overset{\|}{C}}-CH_2-R \longrightarrow R-\overset{O}{\overset{\|}{C}}-OH + HO-\overset{O}{\overset{\|}{C}}-R'$$

(todne ke baad jitne 'C' ka structure ↳ vaisa acid)

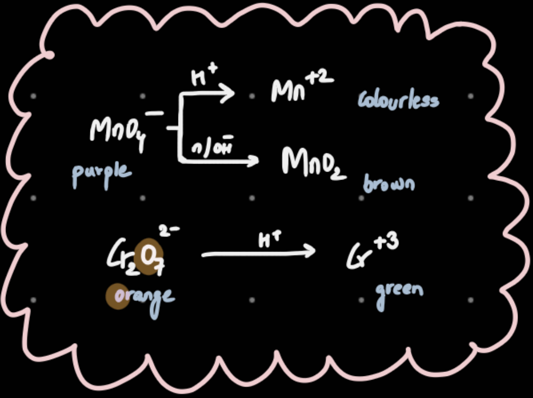
OR

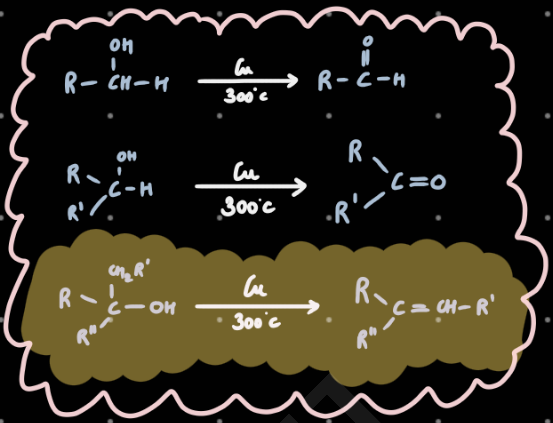


$$R-\overset{OH}{C}=CH-R' \xrightarrow[\text{oxidation (oxidative)}]{\text{vigrous}} R-\overset{OH}{C}=O + HO-\overset{O}{\overset{\|}{C}}-R'$$

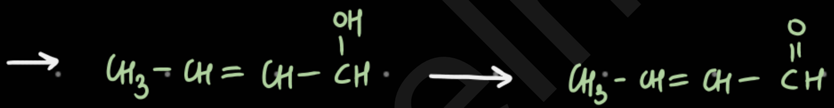
(stable waala enol katega)

3° ⇒ xxx OR → pehle dehydrate karke alkene → Vigorous Oxd.


MnO_4^-
purple
H^+
Mn^{+2}
colourless
n/$\bar{OH}$
MnO_2
brown
$Cr_2O_7^{2-}$
orange
H^+
Cr^{+3}
green


Cu
300°c
Cu
300°c
Cu
300°c

* Jones Oxidation
(H_2CrO_4 + Acetone)



$$CH_3-CH=CH-\overset{OH}{\overset{|}{C}H} \longrightarrow CH_3-CH=CH-\overset{O}{\overset{\|}{C}H}$$

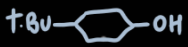
Grey's reagent ← (P.C.C same function)

* $CH_3-\underset{OH}{\underset{|}{CH}}-CH_2CH_2OH$ ← only 1° ⇒ OsO_4

↳ only 2° ⇒ Cl_2/pyridine OR NaOCl in HOAc

* $CH_3\underset{OH}{\underset{|}{CH}}CH_3$ ≫ faster oxidation $CH_3-\underset{OH}{\underset{|}{CD}}-CH_3$ in $Cr_2O_7^{2-}$

$\boxed{E_2}$ ↳ cis > trans

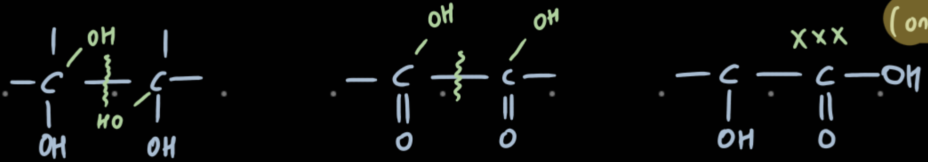


↳ O.S involved : Cr (III)(IV)(V)(VI)

* Ceric Amm. nitrate + Alc
↳ RED colour

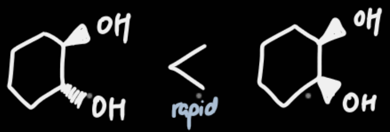
(IV) Per-iodic Acid

①


OH
HO
OH
OH
OH
OH
O
O
xxx
OH
OH
O

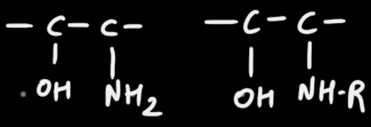
(only Aldehyde / ketone)

②


OH
OH
<
rapid
OH
OH

↳ cyclic intermediate

③ Similar in


OH
NH_2
OH
NH-R

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Facts :

1. $CH_3OH \rightarrow$ **Carbinol**

$CH_3OH \xleftarrow[\text{catalyst}]{} CO + H_2$

2. CH_2CH_2OH → Spirit of wine, **Grain alc.**

↓ fermentation ⤳ Anaerobic

Invertase ✓ zymase ✓

3. Denatured Alc. ⤳ Methylated spirit

↳ $CuSO_4$ → colour

Pyridine → smell